

Parameter Study

Test summary

Table 1: Summary of self-gravity solver parameter values in the study. Each row lists the parameter, its baseline value (used in the reference run), and the test values explored to assess sensitivity.

Parameter	Baseline	Test values		
η (timestep)	0.002	0.0002	0.02	—
MAC	geometric	adaptive	—	—
θ_{cr} (geometric only)	0.7	0.5	—	—
ϵ_{fmm} (FMM tolerance)	0.001	0.0001	0.01	0.1

ϵ_{fmm}

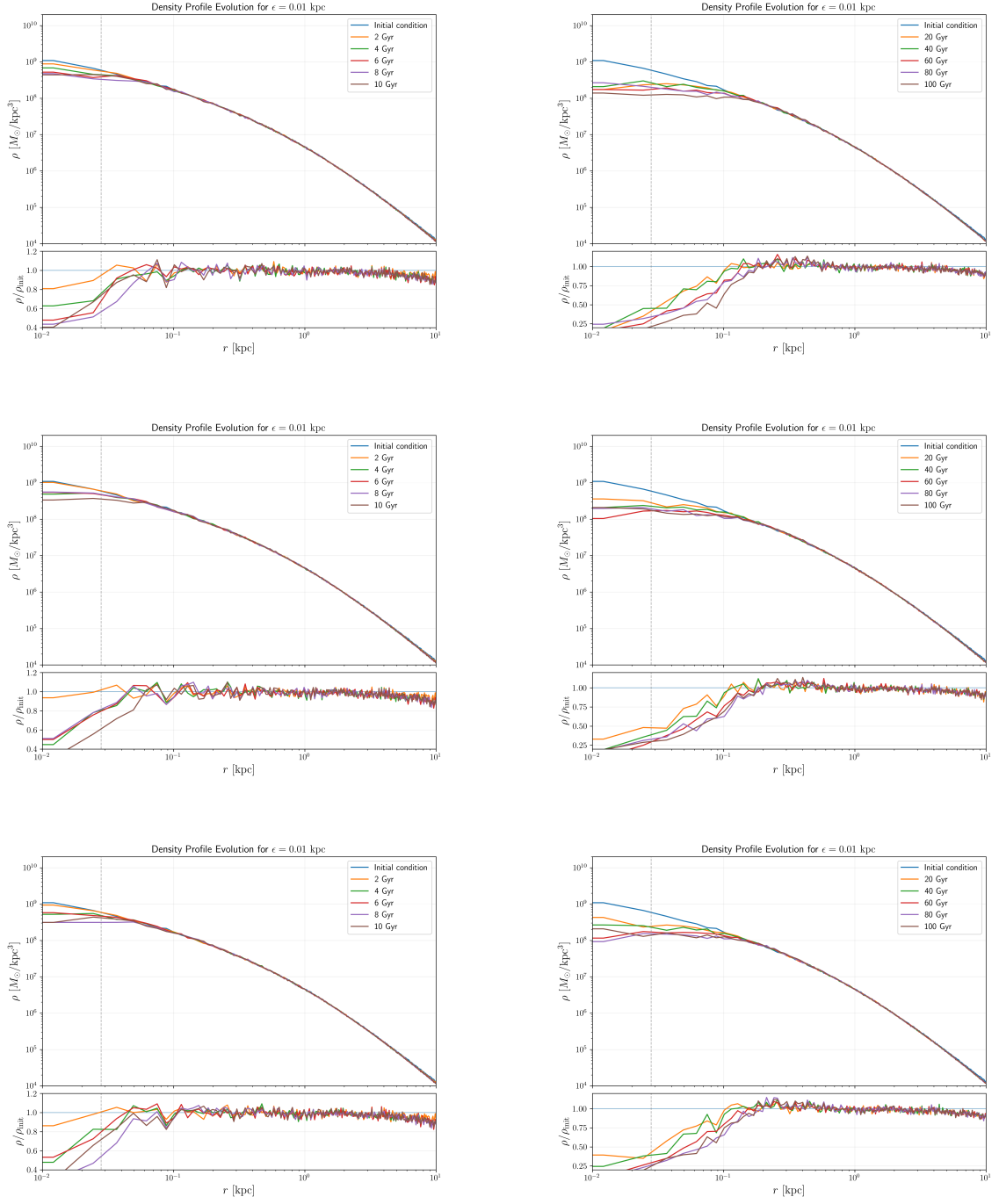


Figure 1: Density evolution for ϵ_{fmm} tests. Left-to-right, top-to-bottom: $\epsilon_{\text{fmm}} = 0.01$ (step 10), 0.01 (step 100), 0.0001 (step 10), 0.0001 (step 100), 0.1 (step 10), 0.1 (step 100).

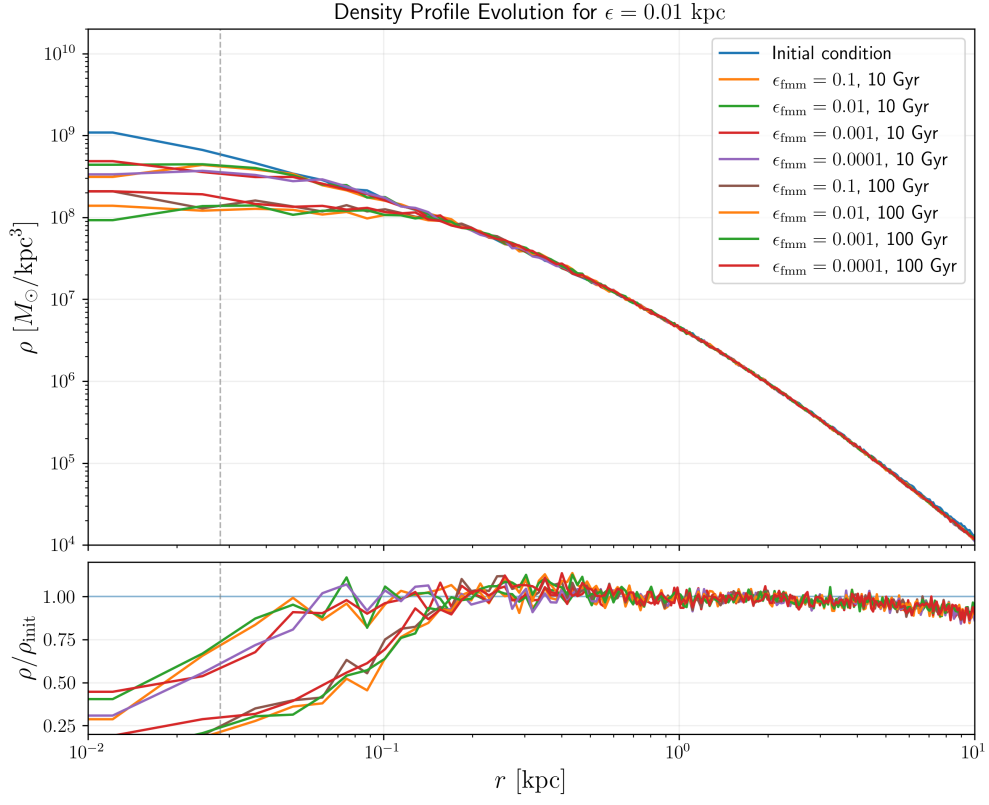


Figure 2: Comparison of ϵ_{fmm} sensitivity: baseline 0.001 versus test values 0.0001, 0.01, 0.1.

MAC

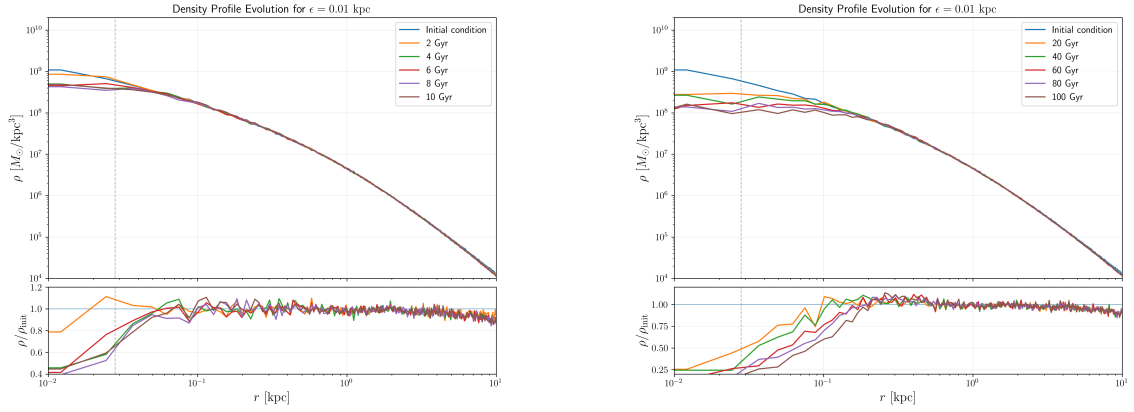


Figure 3: Density evolution for MAC method. Left: adaptive MAC at step 10; Right: adaptive MAC at step 100.

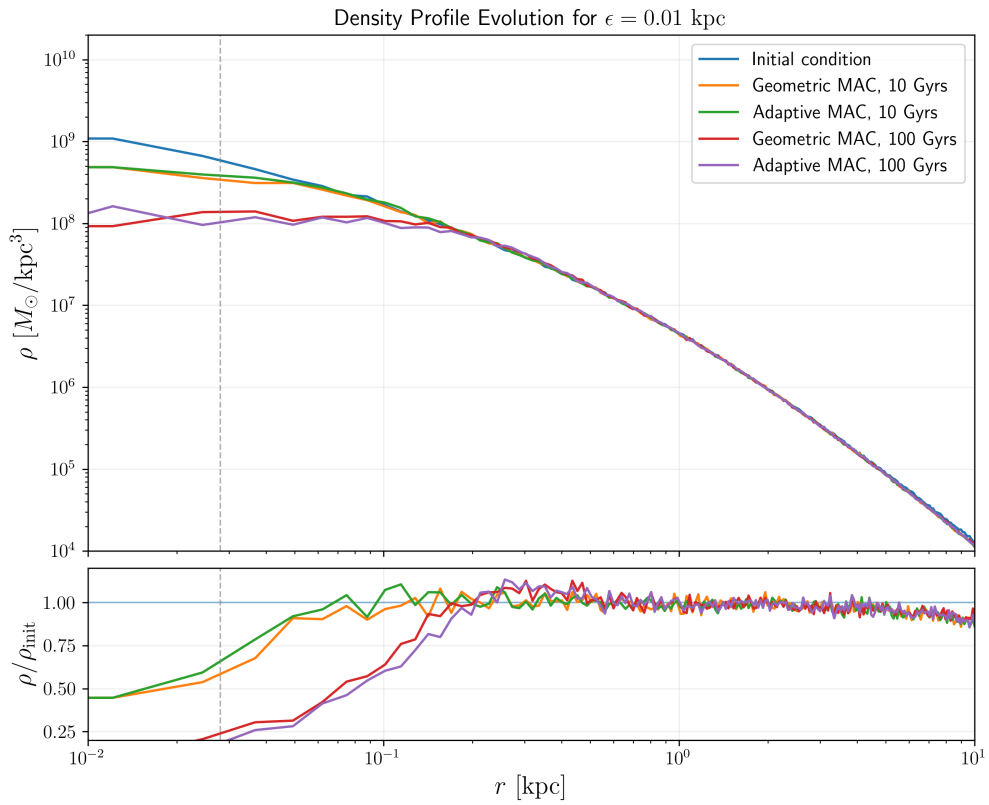


Figure 4: Comparison of MAC method sensitivity: baseline (geometric) versus adaptive MAC.

θ_{cr}

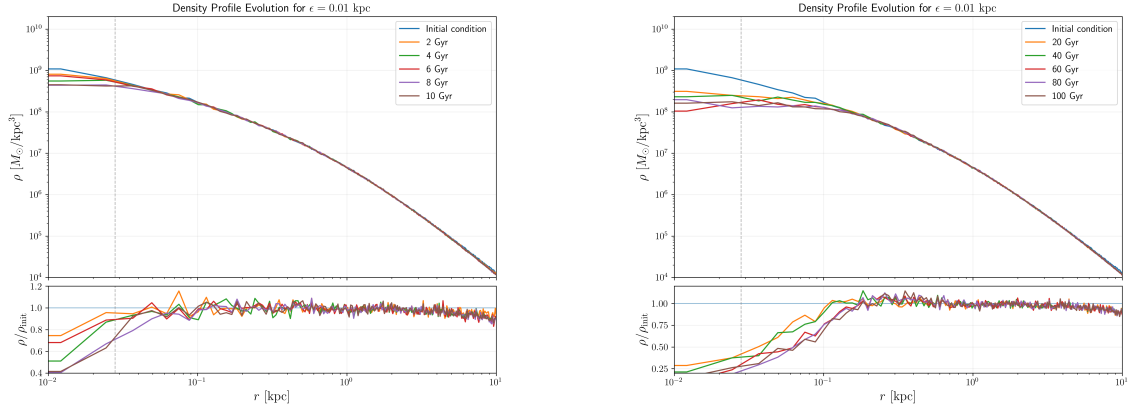


Figure 5: Density evolution for θ_{cr} test. Left: $\theta_{\text{cr}} = 0.5$ at step 10; Right: $\theta_{\text{cr}} = 0.5$ at step 100.

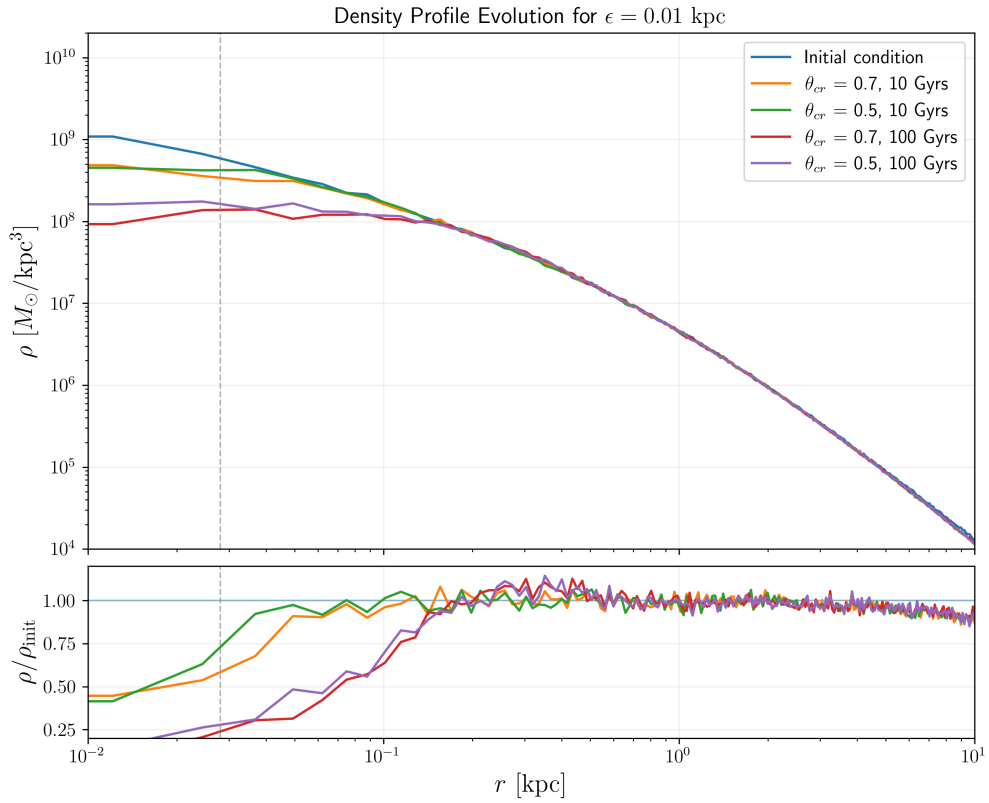


Figure 6: Comparison of θ_{cr} sensitivity: baseline 0.7 versus test value 0.5.

η (timestep)

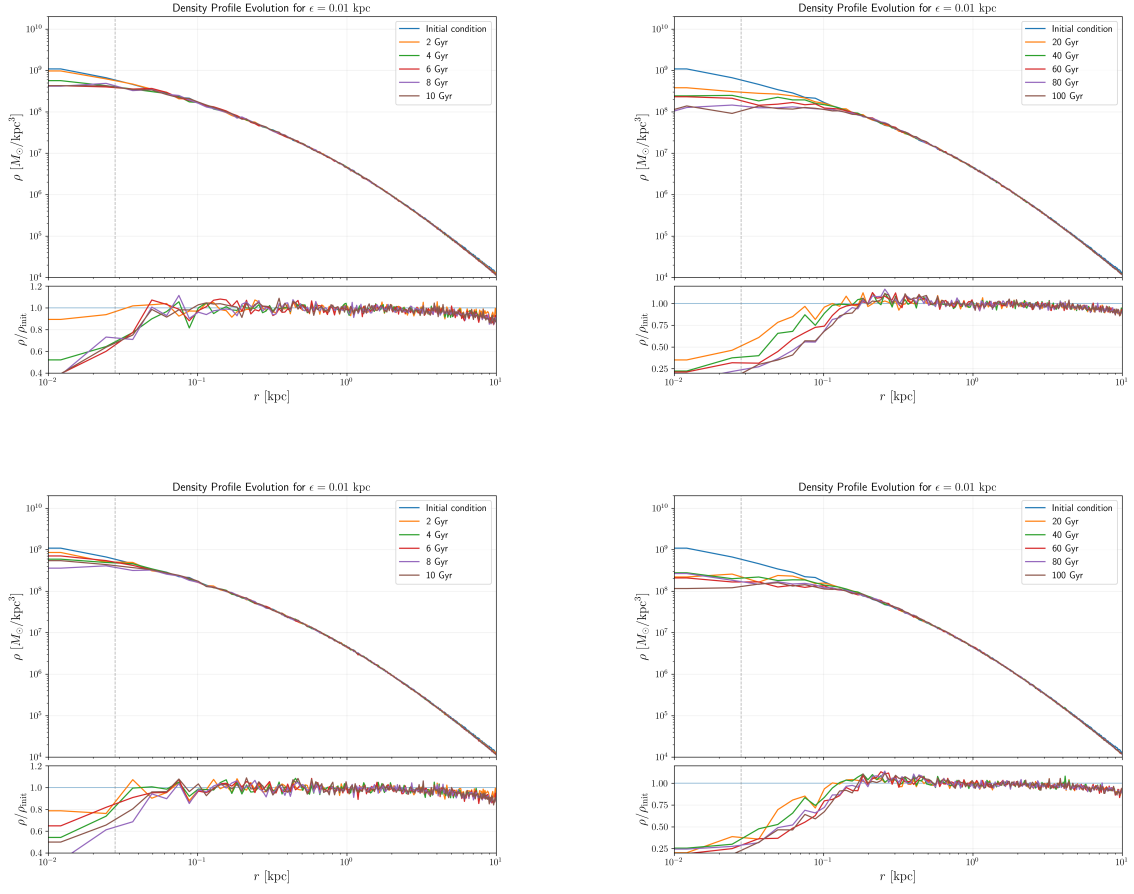


Figure 7: Density evolution for η (timestep) tests. Left-to-right, top-to-bottom: $\eta = 0.02$ (step 10), 0.02 (step 100), 0.0002 (step 10), 0.0002 (step 100).

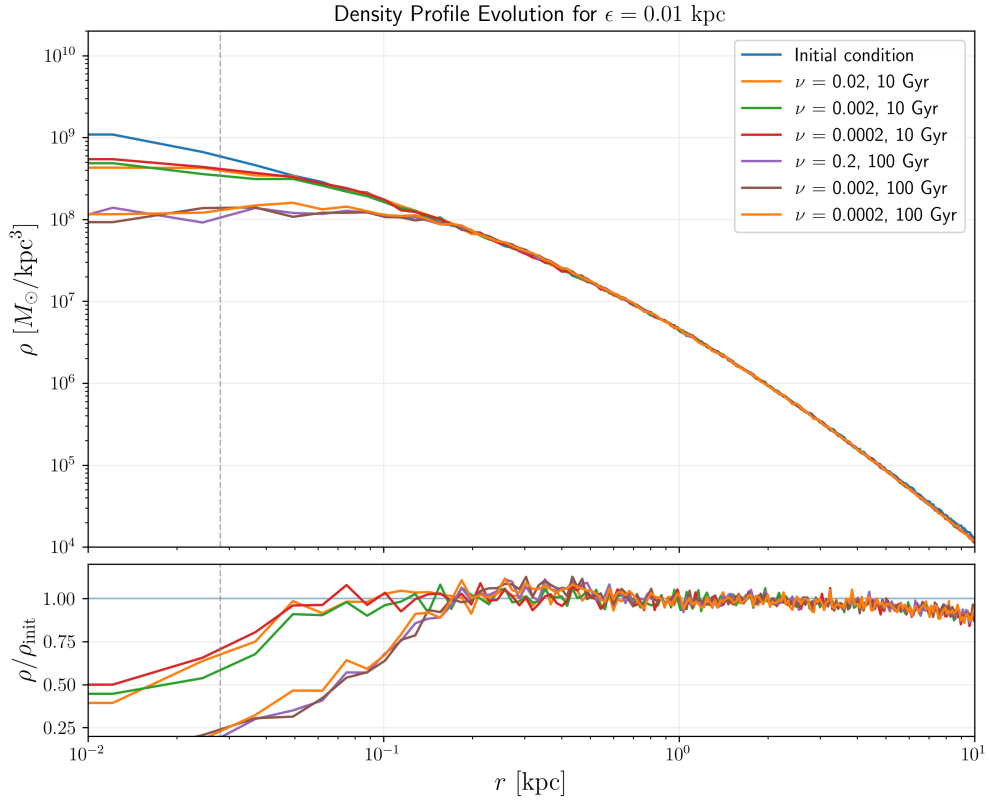


Figure 8: Comparison of η (timestep) sensitivity: baseline 0.002 versus test values 0.0002 and 0.02.